

Bidirectional Mass Spectrometry Modeling Aided by Auxiliary Intermediate Steps From Graph Transformation Tools

Philipp Reiser

TBI Uni Vienna

2026-02-11

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MS-Thesis

Bidirectional Mass Spectrometry Modeling
Aided by Auxiliary Intermediate Steps From
Graph Transformation Tools

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I will talk here about my master thesis project, which I am doing at the University of Vienna in the TBI group.

What is my thesis about?

M ... Molecular structure S ... Mass spectrum

- Formulated mathematically as a function:

$$f : M \rightarrow S$$

- Investigation of the **inverse problem**:

$$f^{-1} : S \rightarrow M$$

- Integration of **chemically studied reaction rules** modeled in **MØD**
- Use of fragmentation trees to support traditional machine-learning-based MS models

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└ Problem

└ What is my thesis about?

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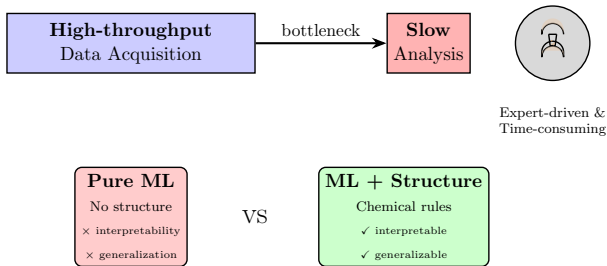
What is this project about?

formulated as mathematically as a function mapping molecular structure to mass spectrum,
and the inverse problem of mapping mass spectrum to molecular structure.

Here we integrate the relationship between the fragments,
which are modeled in MØD which then are used to enhance ML models

Motivation

MS is widely used, but analysis is the challenge



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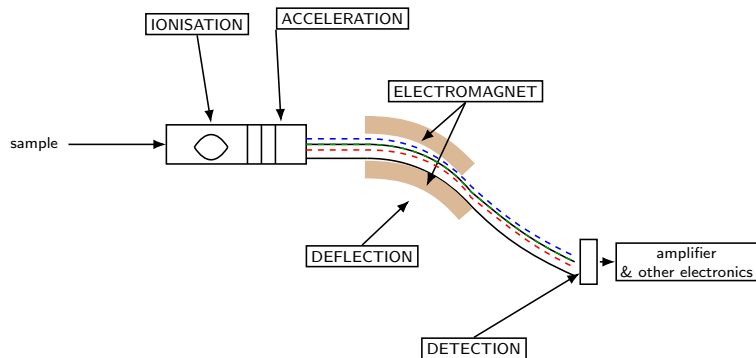
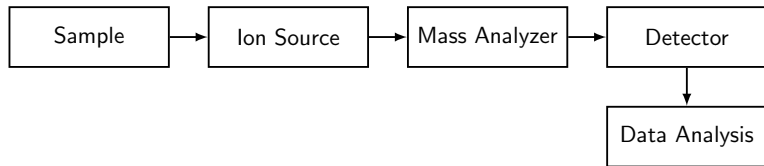
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└ Background
└ Motivation

Motivation



Why would we want to do MS modeling? What is the motivation?
Mass spectrometry is one of the most widely used tools for molecular analysis
As interpreting MS data remains largely expert-driven and time-consuming
the high-through-put in data acquisition does
not get translated through to data analysis, we have a bottleneck.
Also, through incorporating chemical structure, we can enhance machine learning models, in terms of interpretability and generalization

Mass Spectrometer

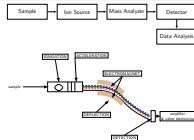


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└ Background

└ Mass Spectrometer

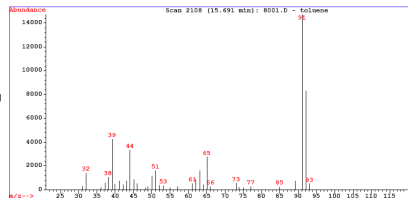
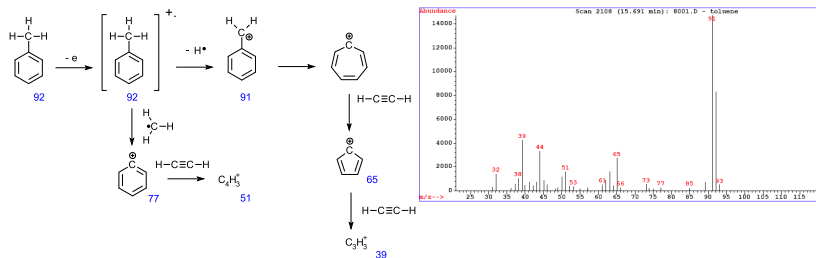
Mass Spectrometer



A quick reminder on how Mass Spec is working.
We have a sample, which is ionized,
then the ions are separated based on their mass-to-charge ratio,
and then detected to produce a spectrum.

Data Sources

- Mass spectrometry data from NIST → SMILES strings & spectra
- Curated EI fragmentation reactions → **MØD** rules



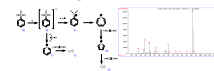
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└ Data

└ Data Sources

Data Sources

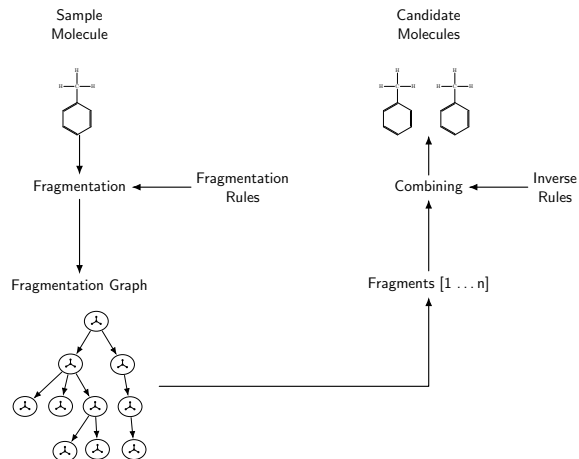
- Mass spectrometry data from NIST → SMILES strings & spectra
- Curated EI fragmentation reactions → **MØD** rules



We have two main data sources:

Mass spectrometry data from NIST,
which gives us the molecular structure in the form of SMILES strings
and spectra,
and curated EI fragmentation reactions, which are modeled as MØD
rules.

Data Generation Overview

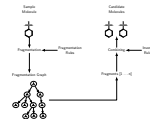


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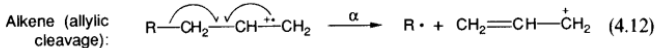
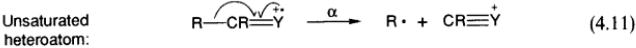
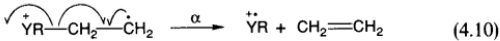
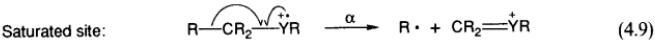
└ Data Generation Overview

Data Generation Overview



a sample molecule is fragmented using the fragmentation rules in MØD, producing a fragmentation graph, from which we can extract the fragments. These fragments are combined with the inverse of the reaction rules to produce candidate precursors for a given spectrum.

Sample Fragmentation Rules



1

2

3

4

5

6

7

8

9

10

```

IMS_4_9 = mod.Rule.fromDFS(
    s =
        "[C]1[C]2[C]3([C]4)[_A+.]5"
        ">>"
        "[C.]1" ". " "[C]2[C]3([C]4){=}_A+.]5",
    name =
        "radical induced (alpha-)cleavage for a saturated site"
        " 4.9"
        "!R1R3R4Y5"
)
```

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└Data

└Sample Fragmentation Rules

Sample Fragmentation Rules

Saturated site:

$$\text{R}-\text{CR}_2-\text{YR} \xrightarrow{\alpha} \text{R}\cdot + \text{CR}_2=\text{YR}^+$$

(4.9)

Unsaturated heteroatom:

$$\text{R}-\text{CR}=\text{Y} \xrightarrow{\alpha} \text{R}\cdot + \text{CR}\equiv\text{Y}^+$$

(4.11)

Alkene (allylic cleavage):

$$\text{R}-\text{CH}_2-\text{CH}^+-\text{CH}_2 \xrightarrow{\alpha} \text{R}\cdot + \text{CH}_2=\text{CH}-\text{CH}_2^+$$

(4.12)

```

1  IMS_4_9 = mod.Rule.fromDFS(
2      s =
3          "[C]1[C]2[C]3([C]4)[_A+.]5"
4          ">>"
5          "[C.]1" ". " "[C]2[C]3([C]4){=}_A+.]5",
6      name =
7          "radical induced (alpha-)cleavage for a saturated site"
8          " 4.9"
9          "!R1R3R4Y5"
10 )
```

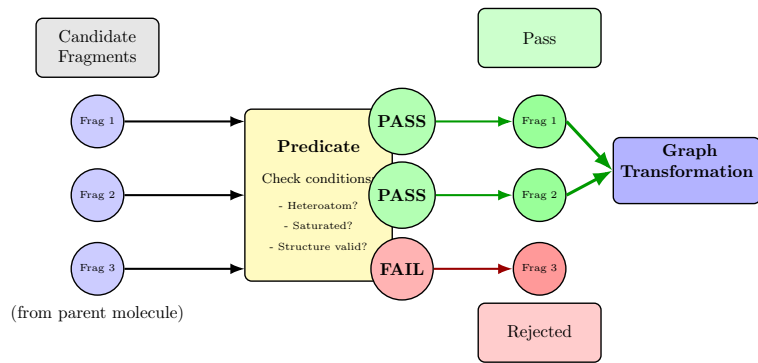
On the top we see example fragmentation rules as stated in the literature,

note that these contain as well as atoms also groups.

On the bottom we see how these rules are represented in the code using a language to encode the groups of atoms.

We use this extended notation in the Graph Transformation step.

MØD - Predicate



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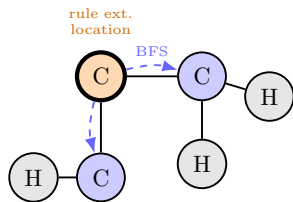
└ MØD - Predicate

MØD - Predicate



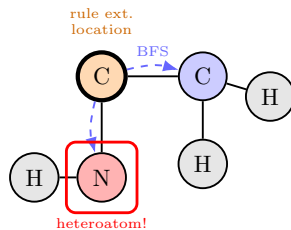
in MØD Predicates act as callbacks during transformation, which I use to filter candidates based on the rule extension. In the following slides I will show examples of the predicates I implemented for filtering based on the the rule extension.

MØD - Predicate - Alkyl Group



VALID: Alkyl Group

BFS neighbors $\subseteq \{H, C\}$ ✓



INVALID: Not Alkyl

BFS neighbors $\not\subseteq \{H, C\}$ ×

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└ Data

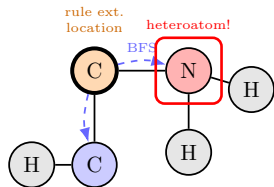
└ MØD - Predicate - Alkyl Group

MØD - Predicate - Alkyl Group



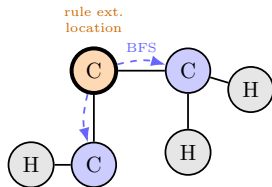
A hydrocarbon substituent is only containing C-H bonds.
Here I collect in a breadth-first manner all neighbors
until they are contained in the match of the original rule

MØD - Predicate - Heteroatom Group



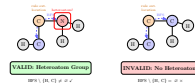
VALID: Heteroatom Group

$$\text{BFS} \setminus \{\text{H}, \text{C}\} \neq \emptyset \checkmark$$



INVALID: No Heteroatom

$$\text{BFS} \setminus \{\text{H}, \text{C}\} = \emptyset \times$$



A valid heteroatom groups must contain at least one non-C/H atom in its neighbors

Here I collect in a breadth-first manner all neighbors that are not matched by the original rule and checks if the resulting set is not disjoint with $\{\text{H}, \text{C}\}$

MØD - Predicate - Saturation

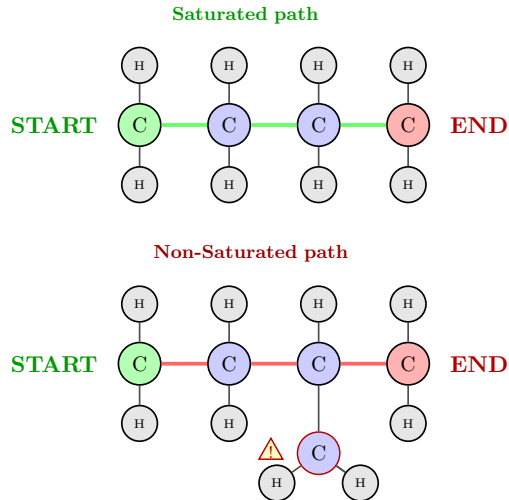
Goal: Find saturated path

Requirements:

- ✓ Only carbon backbone
- ✓ Single bonds only (C—C)
- ✓ Only H side branches

Method:

Search from **start** to **end** following C-C single bonds

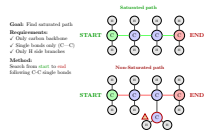


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└ Data

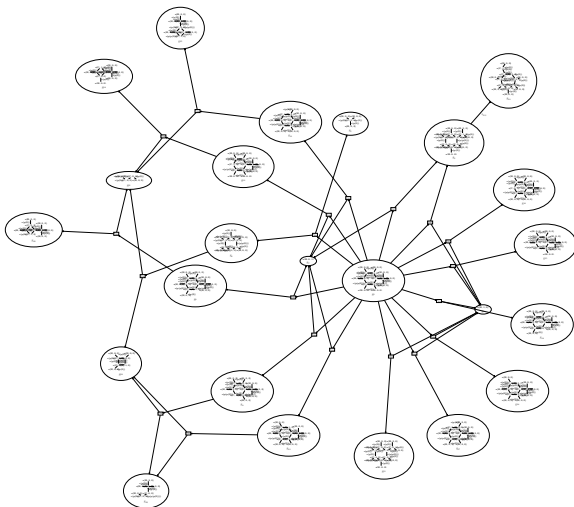
└ MØD - Predicate - Saturation

MØD - Predicate - Saturation



A chain of carbon atoms connected by single bonds, where each carbon may only have hydrogen atoms as side branches

Sample Derivation Graph



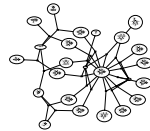
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└ Data

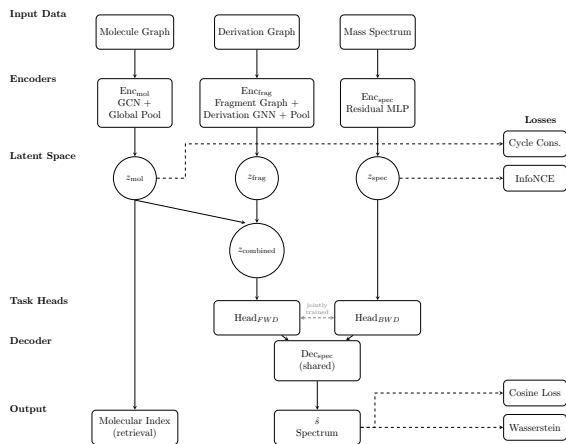
└ Sample Derivation Graph

Sample Derivation Graph



Here we see a the computed output for the fragmentation depicted as a graph,
where the nodes are the fragments and the edges are the reactions that connect them.

Architecture Diagram



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└ ML-Modelling

└ Architecture Diagram

Architecture Diagram



Here we see the architecture of the model I implemented. We have three main inputs: the molecule graph, the fragmentation graph, and the spectrum.

Each of these is encoded into a latent representation using encoders. The latent representations are then combined and fed into task heads for forward and backward prediction.

Finally, we have decoders that produce the predicted fragments and spectrum. In addition, I have

Results

Diverse Model scores:

Best Recall@1: 0.000
Best Recall@5: 0.111
Best Recall@10: 0.259

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└ ML-Modelling
└ Results

Results

Diverse Model scores:

Best Recall@1: 0.000
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For a model trained on a very diverse set of 70 molecules
I achieve a best similarity score for predicting the correct structure
for the top-1 of 0.0% so no molecules was correctly predicted
for the top-5 of 11.1%
for the top-10 of 25.9%

References

- [1] V8rik. *Toluene Fragmentation*. 2008. URL: <https://upload.wikimedia.org/wikipedia/commons/6/6c/TolueneFragmentation.svg> (visited on 01/20/2026).
- [2] Paginazero, Ronhjones. *Toluene Spectra*. 2017. URL: https://upload.wikimedia.org/wikipedia/commons/2/2c/Spettro_di_massa_del_toluene.PNG (visited on 01/20/2026).
- [3] Fred W McLafferty and František Tureček. *Interpretation von Massenspektren*. ger. Spektrum Lehrbuch. Berlin [u.a.]: Spektrum Akad. Verl., 1995. ISBN: 9783642398483.

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└ End

└ References

these are my sources and references

References

- [1] V8rik. *Toluene Fragmentation*. 2008. URL: <https://upload.wikimedia.org/wikipedia/commons/6/6c/TolueneFragmentation.svg> (visited on 01/20/2026).
- [2] Paginazero, Ronhjones. *Toluene Spectra*. 2017. URL: https://upload.wikimedia.org/wikipedia/commons/2/2c/Spettro_di_massa_del_toluene.PNG (visited on 01/20/2026).
- [3] Fred W McLafferty and František Tureček. *Interpretation von Massenspektren*. ger. Spektrum Lehrbuch. Berlin [u.a.]: Spektrum Akad. Verl., 1995. ISBN: 9783642398483.

Problem ○	Background ○○	Data ○○○○○○○○	ML-Modelling ○○	End ○●
Discussion				
Questions?				

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└ End

└ Discussion

Discussion

Questions?

thank you for your attention
Do you have questions?